HW4_Part1_CSchuler

[1]
0s

```
import pandas as pd
import matplotlib.pyplot as plt
```

[2]
25s

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

Q1 Please use Pandas to read olympic_medals.csv and use parallel_categories function from plotly.express to visualize proportions of medal type for each gender from since year 2000. Please see the example in the Python notebook we walked through in the class.

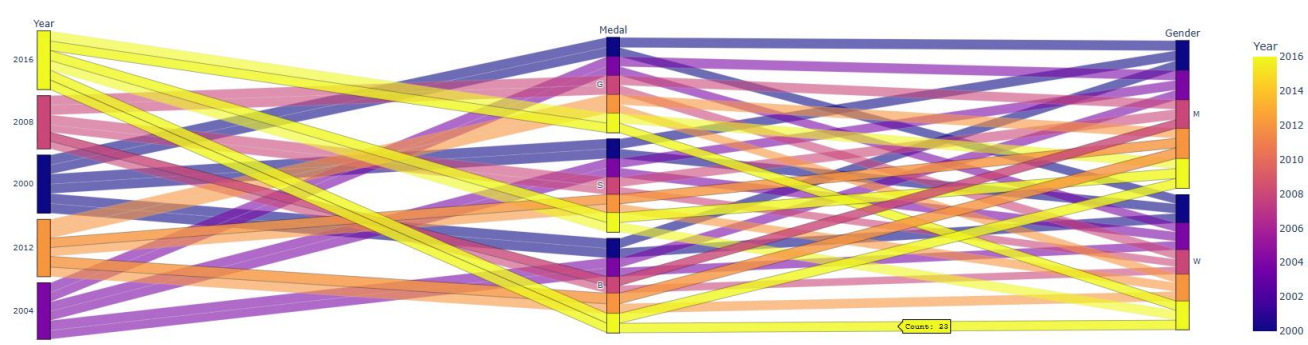
[3]
3s

```
#please use this cell to read and select your data

df = pd.read_csv('/content/drive/MyDrive/DATA/olympic_medals.csv')
print(df.head())
```

```
  Gender       Event Location  Year Medal                  Name Nationality  \
0      M  10000M Men      Rio  2016     G          Mohamed FARAH         GBR
1      M  10000M Men      Rio  2016     S  Paul Kipngetich TANUI         KEN
2      M  10000M Men      Rio  2016     B           Tamirat TOLA         ETH
3      M  10000M Men  Beijing  2008     G        Kenenisa BEKELE         ETH
4      M  10000M Men  Beijing  2008     S         Sileshi SIHINE         ETH

     Result
0  25:05.17
1  27:05.64
2  27:06.26
3  27:01.17
4  27:02.77
```

[4]
6s

```
#Please use this cell to create your your figure. Please use Year column to color your graph.

import plotly.express as px

df_filter = df[df['Year'] >= 2000]

px.parallel_categories(df_filter, dimensions=['Year', 'Medal', 'Gender'], color="Year", color_continuous_scale=px.colors.sequential.Plasma)
```

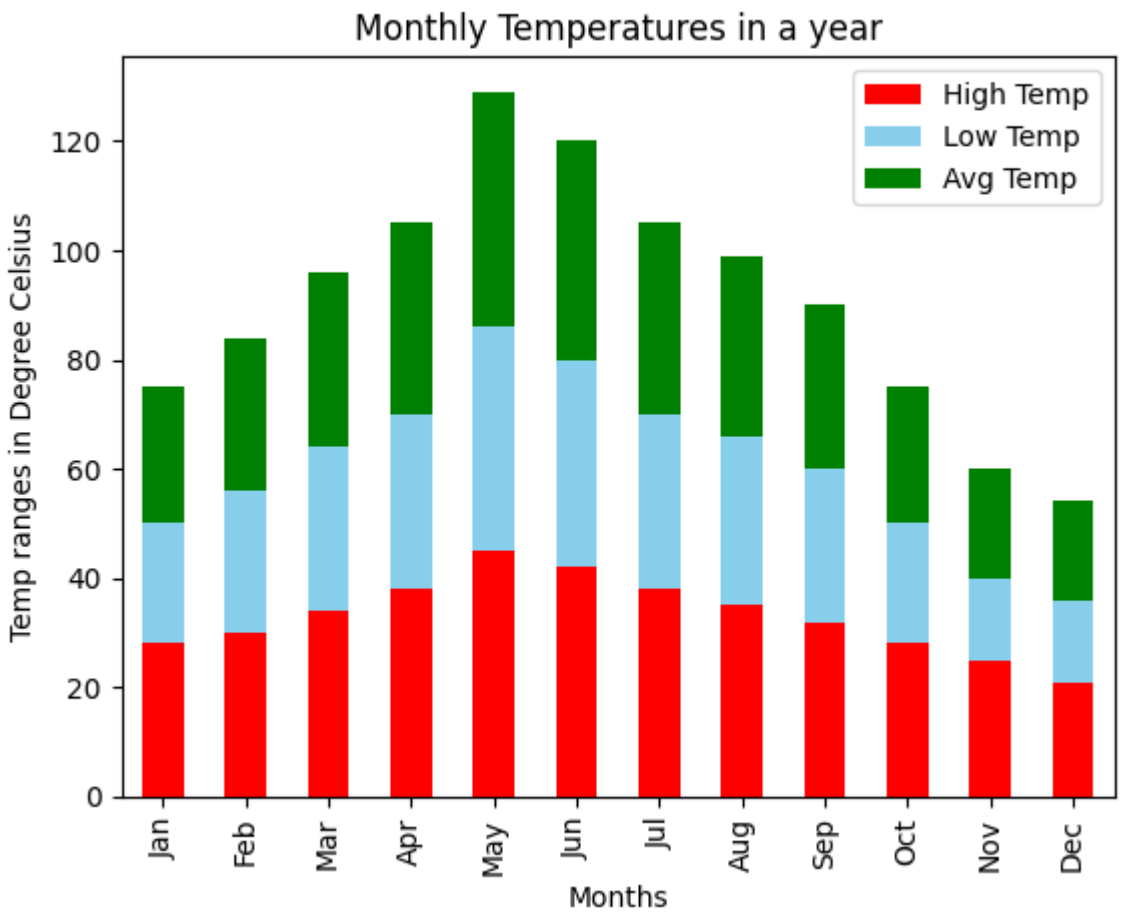


Q2 Please inspect the code below and observe how values are plotted by running it. Then, read the 2016elections.csv from the DATA folder and select rows for AR, MI, CA, and WI. Then, utilize stacked bar plot, to stack vote percentages for Trump, Clinton, Johnson, and Others. Please see 'pct_clinton', 'pct_trump', 'pct_johnson', 'pct_other' columns. Make sure that your x tick labels are those four states above.

[5]
✓ 1s

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# create DataFrame
df = pd.DataFrame({'High Temp': [28, 30, 34, 38, 45, 42,
                                 38, 35, 32, 28, 25, 21],
                   'Low Temp': [22, 26, 30, 32, 41, 38,
                                32, 31, 28, 22, 15, 15],
                   'Avg Temp': [25, 28, 32, 35, 43, 40,
                                35, 33, 30, 25, 20, 18]},
                  index=['Jan', 'Feb', 'Mar', 'Apr', 'May',
                         'Jun', 'Jul', 'Aug', 'Sep', 'Oct',
                         'Nov', 'Dec'])


# create stacked bar chart for monthly temperatures
df.plot(kind='bar', stacked=True, color=['red', 'skyblue', 'green'])

# labels for x & y axis
plt.xlabel('Months')
plt.ylabel('Temp ranges in Degree Celsius')

# title of plot
plt.title('Monthly Temperatures in a year')
```

```
Text(0.5, 1.0, 'Monthly Temperatures in a year')
```



[6]

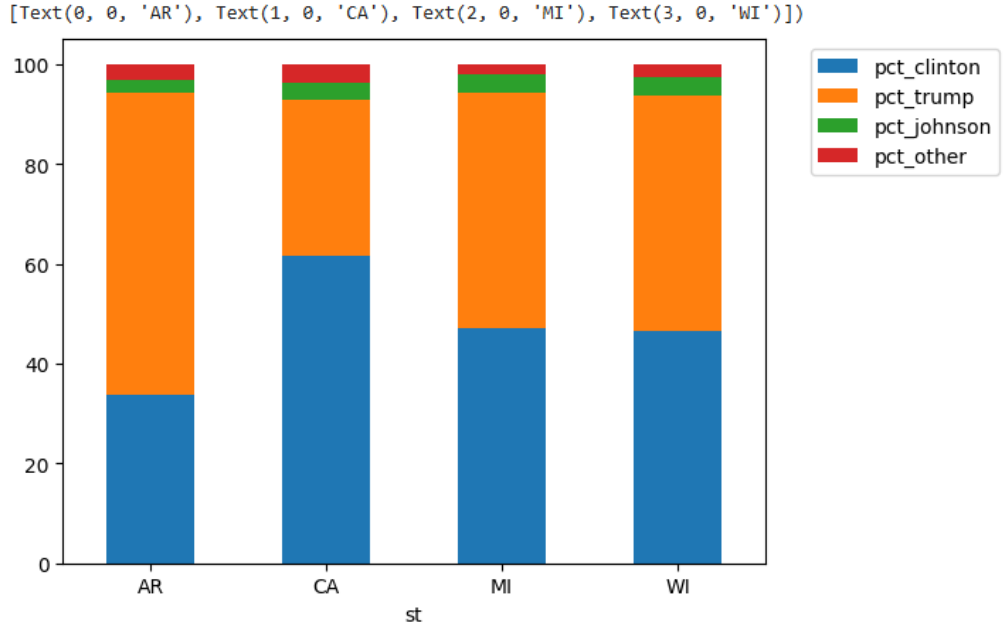
```python
elections = pd.read_csv('/content/drive/MyDrive/DATA/2016elections.csv')
print(elections.head())
```

```
         state  st  fips  total_vote  vote_margin   winner       party  \
0      Alabama  AL     1     2123372       588708    Trump  Republican
1       Alaska  AK     2      318608        46933    Trump  Republican
2      Arizona  AZ     4     2604657        91234    Trump  Republican
3     Arkansas  AR     5     1130635       304378    Trump  Republican
4   California  CA     6    14237893      4269978  Clinton  Democratic

   pct_margin  r_points  d_points  ...  pct_johnson  pct_other  clinton_vote  \
0      0.2773     27.72    -27.72  ...         2.09       1.46        729547
1      0.1473     14.73    -14.73  ...         5.88       6.29        116454
2      0.0350      3.50     -3.50  ...         4.08       3.25       1161167
3      0.2692     26.92    -26.92  ...         2.64       3.13        380494
4      0.2999    -29.99     29.99  ...         3.36       3.66       8753792

   trump_vote  johnson_vote  other_vote  ev_dem  ev_rep  ev_oth  census
0     1318255         44467       31103       0       9       0   South
1      163387         18725       20042       0       3       0    West
2     1252401        106327       84762       0      11       0    West
3      684872         29829       35440       0       6       0   South
4     4483814        478500      521787      55       0       0    West

[5 rows x 22 columns]
```

[7]

```python
#You can use this cell to write your code. It is doable at most 4 lines of code.

elections_filtered = elections[elections['st'].isin(['AR', 'MI', 'CA', 'WI'])].set_index('st')
elections_filtered[['pct_clinton', 'pct_trump', 'pct_johnson', 'pct_other']].plot(kind='bar', stacked=True, legend=True)
plt.legend(loc='upper left', bbox_to_anchor=(1.05, 1))
plt.xticks(rotation=0)
```

```
(array([0, 1, 2, 3]),
 [Text(0, 0, 'AR'), Text(1, 0, 'CA'), Text(2, 0, 'MI'), Text(3, 0, 'WI')])
```



HW4_Part2_CSchuler

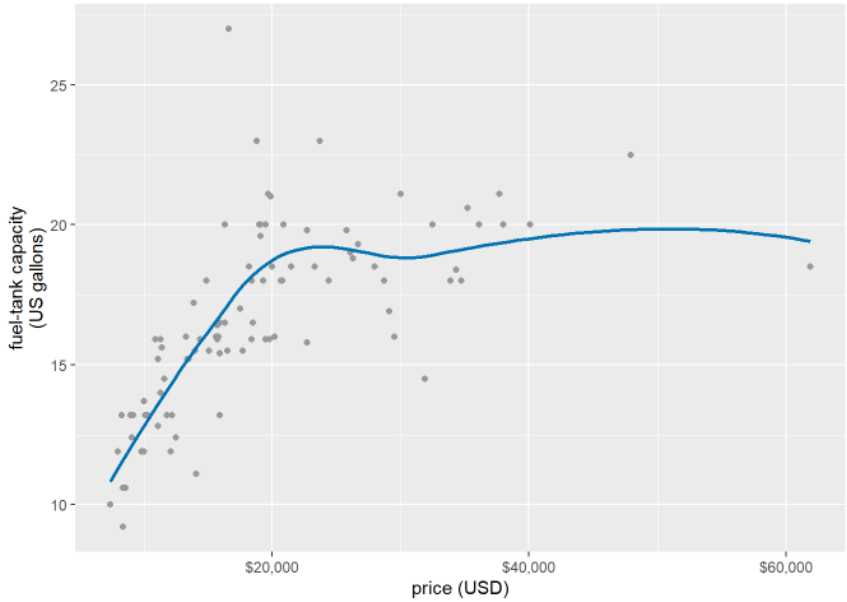
2026-03-05

```
library(tidyverse)
```

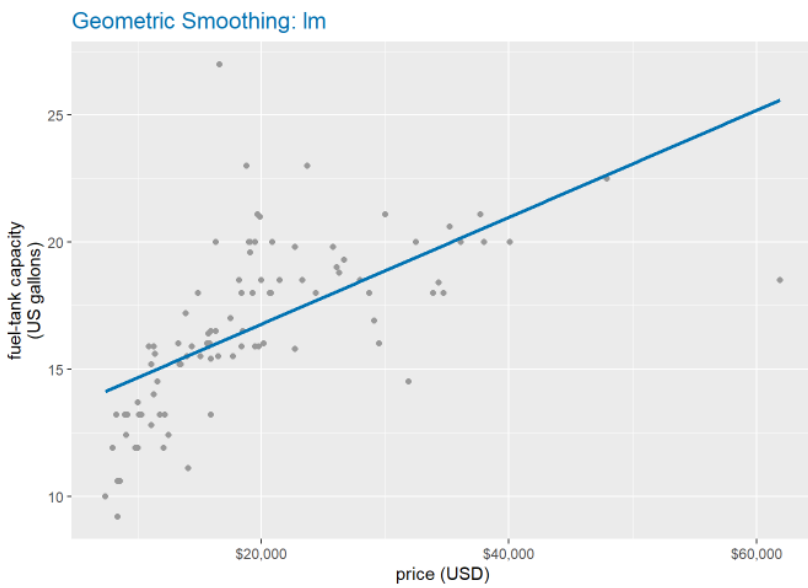
```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✓ dplyr     1.2.0     ✓ readr     2.2.0
## ✓ forcats   1.0.1     ✓ stringr   1.6.0
## ✓ ggplot2   4.0.2     ✓ tibble    3.3.1
## ✓ lubridate 1.9.5     ✓ tidyr     1.3.2
## ✓ purrr     1.2.1
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✕ dplyr::filter() masks stats::filter()
## ✕ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
library(ggridges)
library(lubridate)
library(ggrepel)
library(colorspace)
```

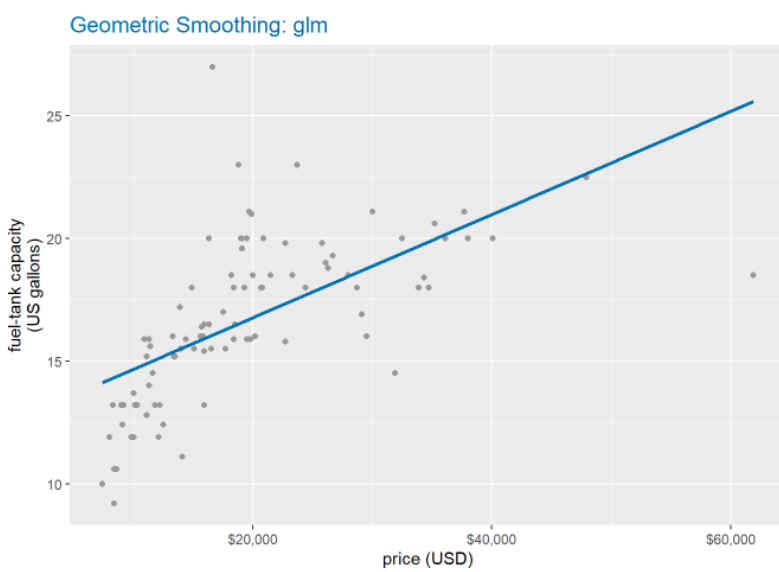
```
#4
cars93 <- MASS::Cars93
ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
geom_point(color = "grey60") +
geom_smooth(se = FALSE, method = "loess", formula = y ~ x, color = "#0072B2") +
scale_x_continuous(
name = "price (USD)",
breaks = c(20, 40, 60),
labels = c("$20,000", "$40,000", "$60,000")
) +
scale_y_continuous(name = "fuel-tank capacity\n(US gallons)")
```



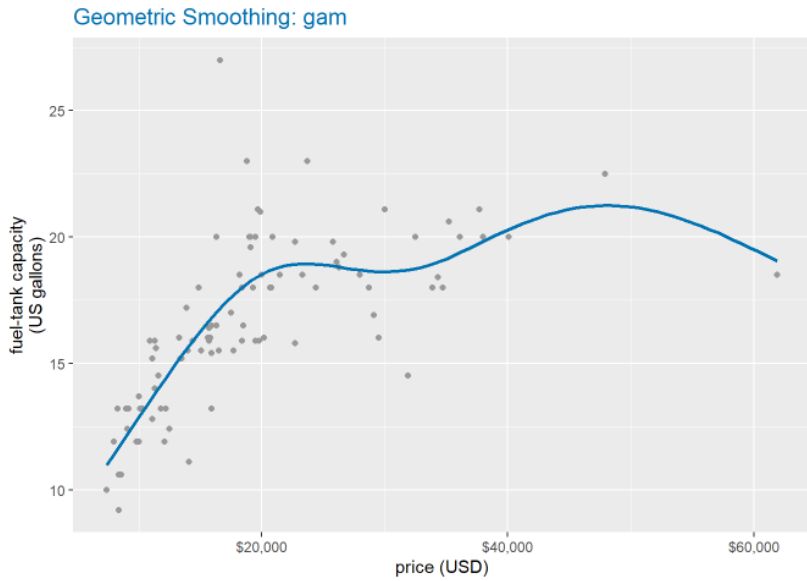
```r
#4a,d,e - lm
ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
geom_point(color = "grey60") +
geom_smooth(se = FALSE, method = "lm", formula = y ~ x, color = "#0072B2") +
scale_x_continuous(
name = "price (USD)",
breaks = c(20, 40, 60),
labels = c("$20,000", "$40,000", "$60,000")
) +
scale_y_continuous(name = "fuel-tank capacity\n(US gallons)") +
labs(title = "Geometric Smoothing: lm") +
theme(
  plot.title = element_text(size = 14, color = "#0072B2")
)
```



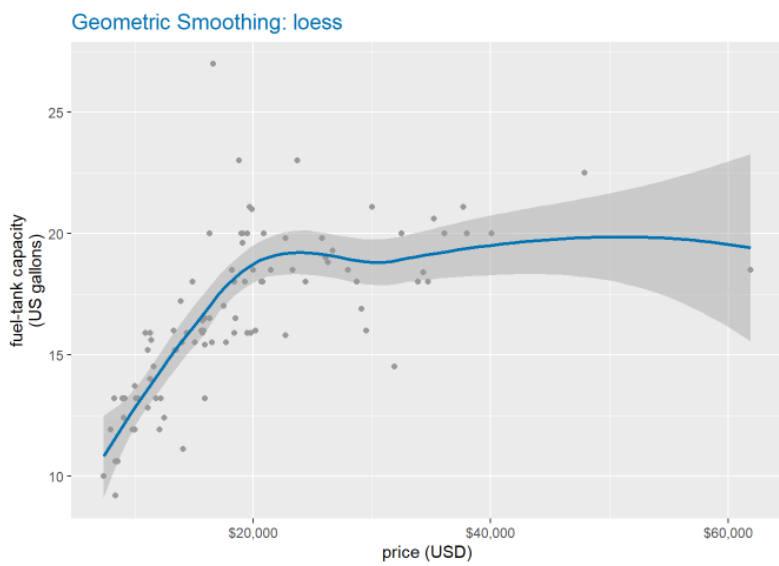

```r
#4a,d,e - glm
ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
geom_point(color = "grey60") +
geom_smooth(se = FALSE, method = "glm", formula = y ~ x, color = "#0072B2") +
scale_x_continuous(
name = "price (USD)",
breaks = c(20, 40, 60),
labels = c("$20,000", "$40,000", "$60,000")
) +
scale_y_continuous(name = "fuel-tank capacity\n(US gallons)")+
labs(title = "Geometric Smoothing: glm") +
theme(
  plot.title = element_text(size = 14, color = "#0072B2"))
```



```r
#4a,d,e - gam
ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
geom_point(color = "grey60") +
geom_smooth(se = FALSE, method = "gam", formula = y ~ s(x), color = "#0072B2") +
scale_x_continuous(
name = "price (USD)",
breaks = c(20, 40, 60),
labels = c("$20,000", "$40,000", "$60,000")
) +
scale_y_continuous(name = "fuel-tank capacity\n(US gallons)")+
labs(title = "Geometric Smoothing: gam") +
theme(
  plot.title = element_text(size = 14, color = "#0072B2"))
```




```r
#4b,d,e
cars93 <- MASS::Cars93
ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
geom_point(color = "grey60") +
geom_smooth(se = TRUE, method = "loess", formula = y ~ x, color = "#0072B2") +
scale_x_continuous(
name = "price (USD)",
breaks = c(20, 40, 60),
labels = c("$20,000", "$40,000", "$60,000")
) +
scale_y_continuous(name = "fuel-tank capacity\n(US gallons)")+
labs(title = "Geometric Smoothing: loess") +
theme(
  plot.title = element_text(size = 14, color = "#0072B2"))
```



```r
#4b,d,e - lm
ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
geom_point(color = "grey60") +
geom_smooth(se = TRUE, method = "lm", formula = y ~ x, color = "#0072B2") +
scale_x_continuous(
name = "price (USD)",
breaks = c(20, 40, 60),
labels = c("$20,000", "$40,000", "$60,000")
) +
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labs(title = "Geometric Smoothing: lm") +
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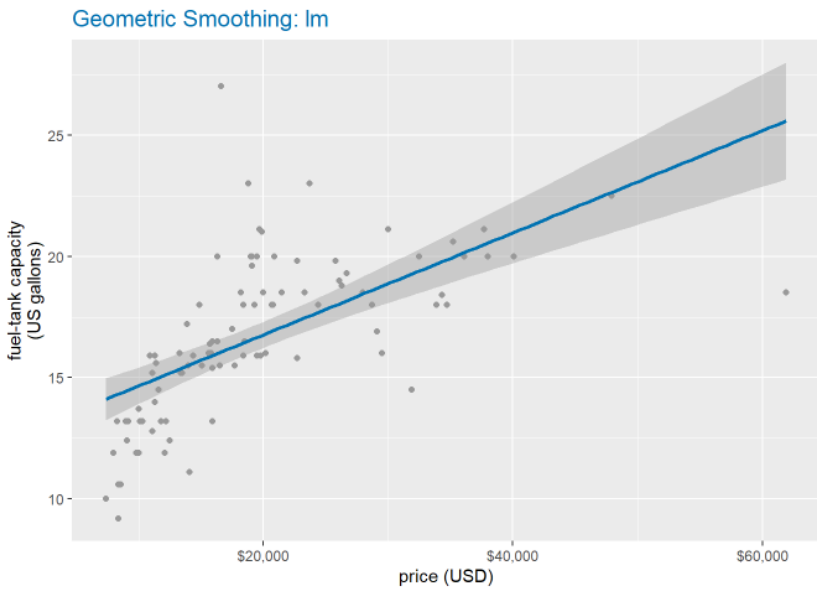
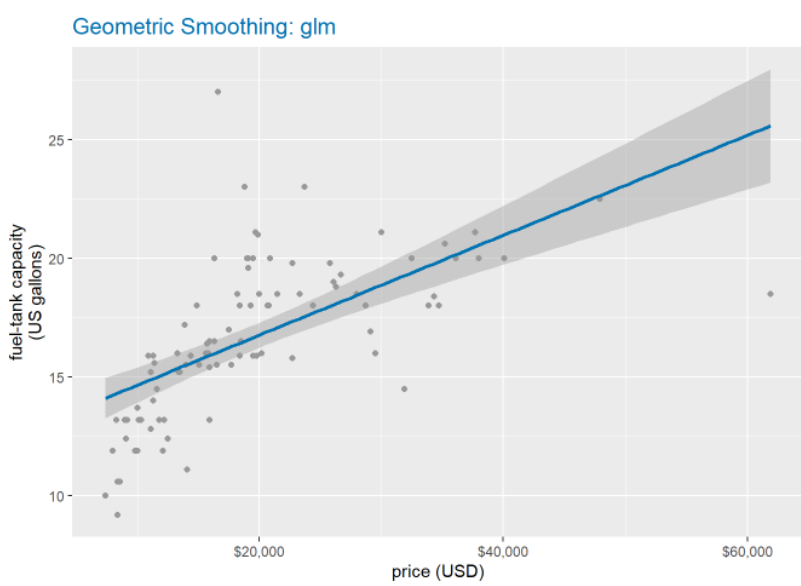
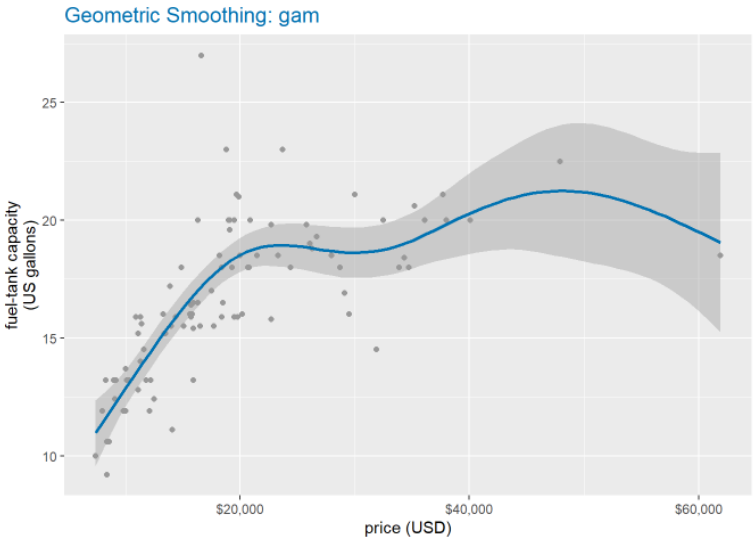


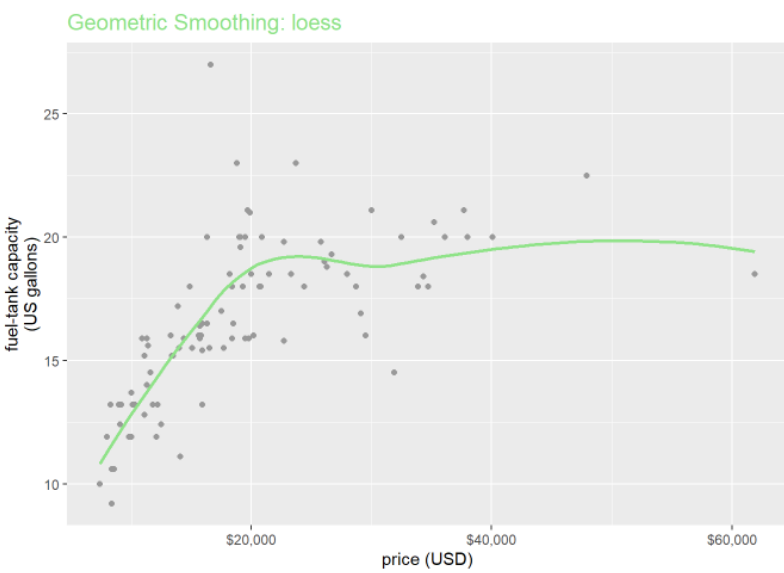

```r
#4b,d,e - glm
ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
geom_point(color = "grey60") +
geom_smooth(se = TRUE, method = "glm", formula = y ~ x, color = "#0072B2") +
scale_x_continuous(
name = "price (USD)",
breaks = c(20, 40, 60),
labels = c("$20,000", "$40,000", "$60,000")
) +
scale_y_continuous(name = "fuel-tank capacity\n(US gallons)")+
labs(title = "Geometric Smoothing: glm") +
theme(
  plot.title = element_text(size = 14, color = "#0072B2"))
```



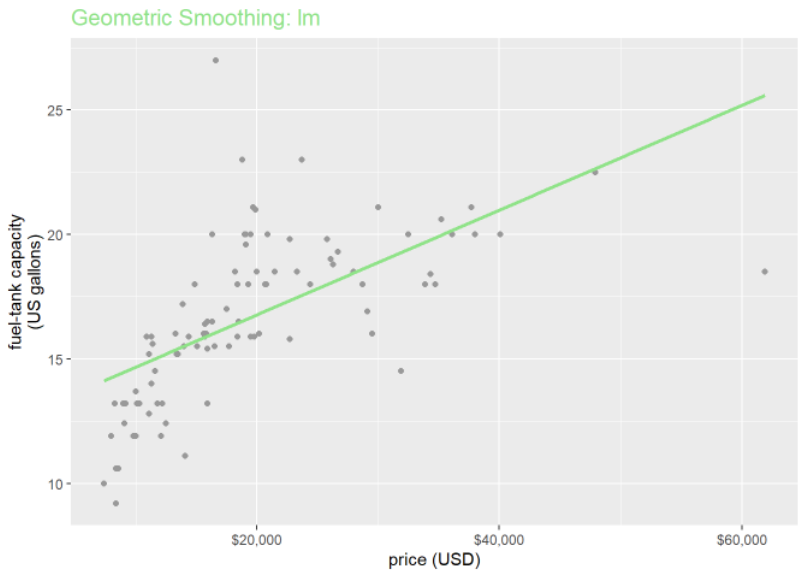
```r
#4b,d,e - gam
ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
geom_point(color = "grey60") +
geom_smooth(se = TRUE, method = "gam", formula = y ~ s(x), color = "#0072B2") +
scale_x_continuous(
name = "price (USD)",
breaks = c(20, 40, 60),
labels = c("$20,000", "$40,000", "$60,000")
) +
scale_y_continuous(name = "fuel-tank capacity\n(US gallons)")+
labs(title = "Geometric Smoothing: gam") +
theme(
  plot.title = element_text(size = 14, color = "#0072B2"))
```



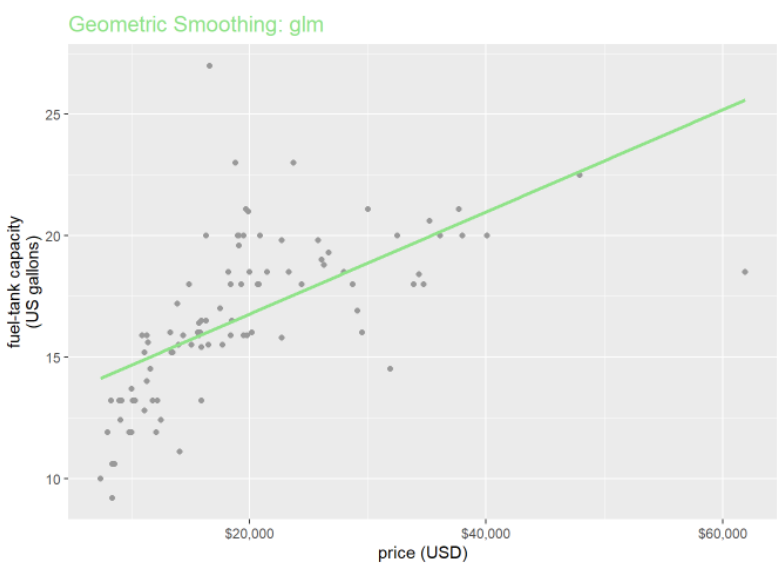

```r
#4c,d,e, 8fe388
ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
geom_point(color = "grey60") +
geom_smooth(se = FALSE, method = "loess", formula = y ~ x, color = "#8fe388") +
scale_x_continuous(
name = "price (USD)",
breaks = c(20, 40, 60),
labels = c("$20,000", "$40,000", "$60,000")
) +
scale_y_continuous(name = "fuel-tank capacity\n(US gallons)")+
labs(title = "Geometric Smoothing: loess") +
theme(
  plot.title = element_text(size = 14, color = "#8fe388"))
```



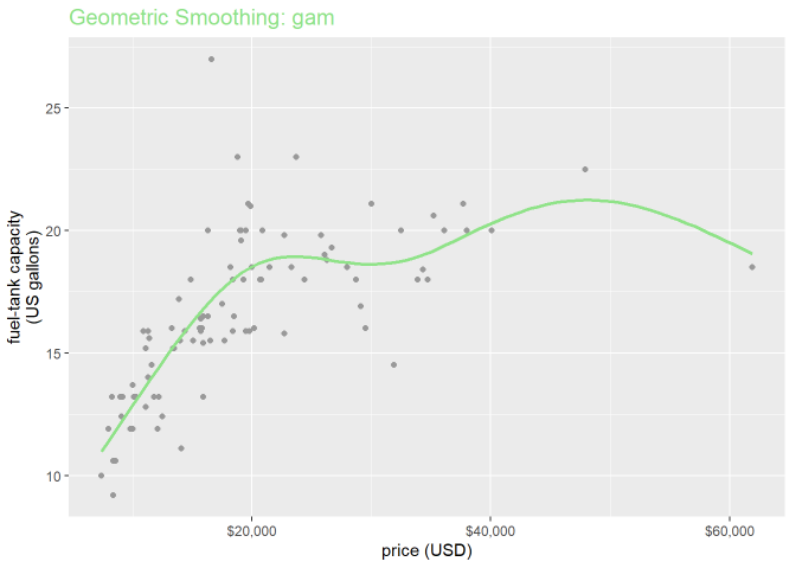
```r
#4c,d,e - lm, 8fe388
ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
geom_point(color = "grey60") +
geom_smooth(se = FALSE, method = "lm", formula = y ~ x, color = "#8fe388") +
scale_x_continuous(
name = "price (USD)",
breaks = c(20, 40, 60),
labels = c("$20,000", "$40,000", "$60,000")
) +
scale_y_continuous(name = "fuel-tank capacity\n(US gallons)")+
labs(title = "Geometric Smoothing: lm") +
theme(
  plot.title = element_text(size = 14, color = "#8fe388"))
```



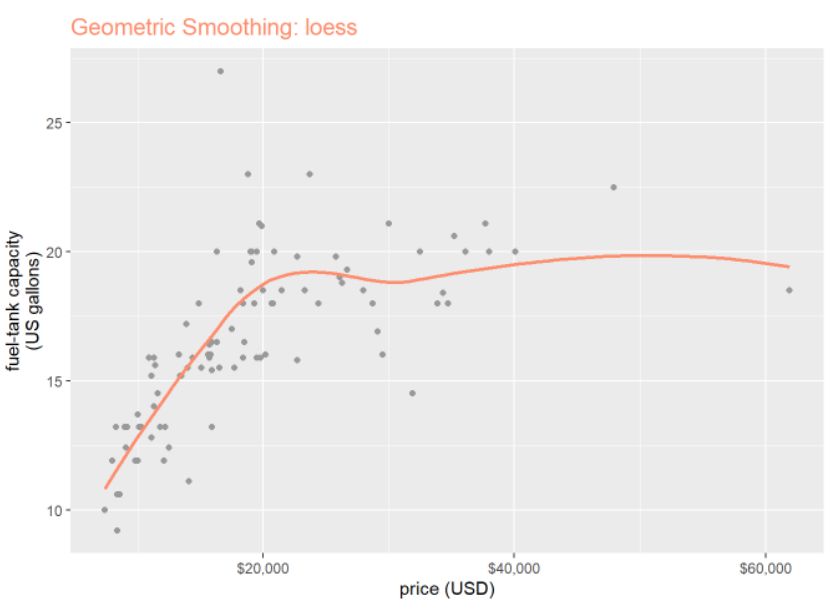

```r
#4c,d,e - glm, 8fe388
ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
geom_point(color = "grey60") +
geom_smooth(se = FALSE, method = "glm", formula = y ~ x, color = "#8fe388") +
scale_x_continuous(
name = "price (USD)",
breaks = c(20, 40, 60),
labels = c("$20,000", "$40,000", "$60,000")
) +
scale_y_continuous(name = "fuel-tank capacity\n(US gallons)")+
labs(title = "Geometric Smoothing: glm") +
theme(
  plot.title = element_text(size = 14, color = "#8fe388"))
```



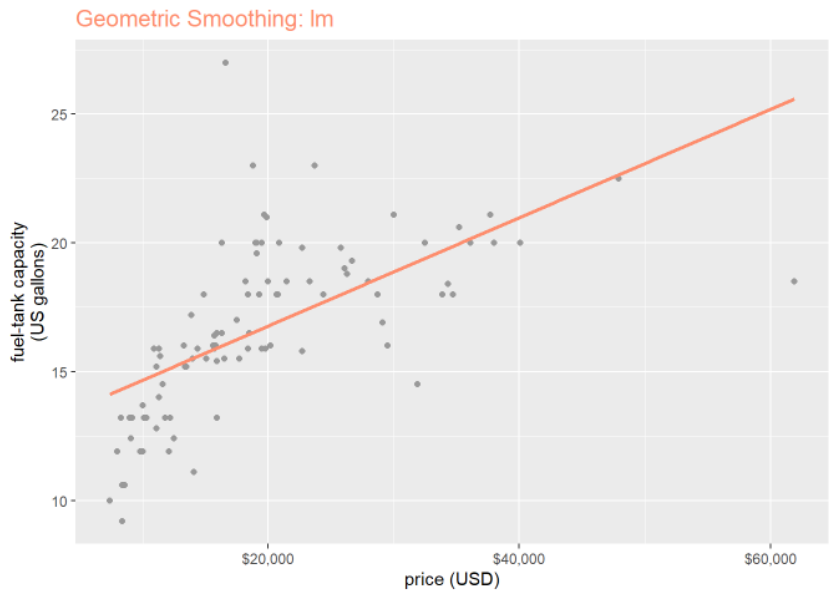
```r
#4c,d,e - gam, 8fe388
ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
geom_point(color = "grey60") +
geom_smooth(se = FALSE, method = "gam", formula = y ~ s(x), color = "#8fe388") +
scale_x_continuous(
name = "price (USD)",
breaks = c(20, 40, 60),
labels = c("$20,000", "$40,000", "$60,000")
) +
scale_y_continuous(name = "fuel-tank capacity\n(US gallons)")+
labs(title = "Geometric Smoothing: gam") +
theme(
  plot.title = element_text(size = 14, color = "#8fe388"))
```




```r
#4c,d,e, fe8d6d
ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
geom_point(color = "grey60") +
geom_smooth(se = FALSE, method = "loess", formula = y ~ x, color = "#fe8d6d") +
scale_x_continuous(
name = "price (USD)",
breaks = c(20, 40, 60),
labels = c("$20,000", "$40,000", "$60,000")
) +
scale_y_continuous(name = "fuel-tank capacity\n(US gallons)")+
labs(title = "Geometric Smoothing: loess") +
theme(
  plot.title = element_text(size = 14, color = "#fe8d6d"))
```

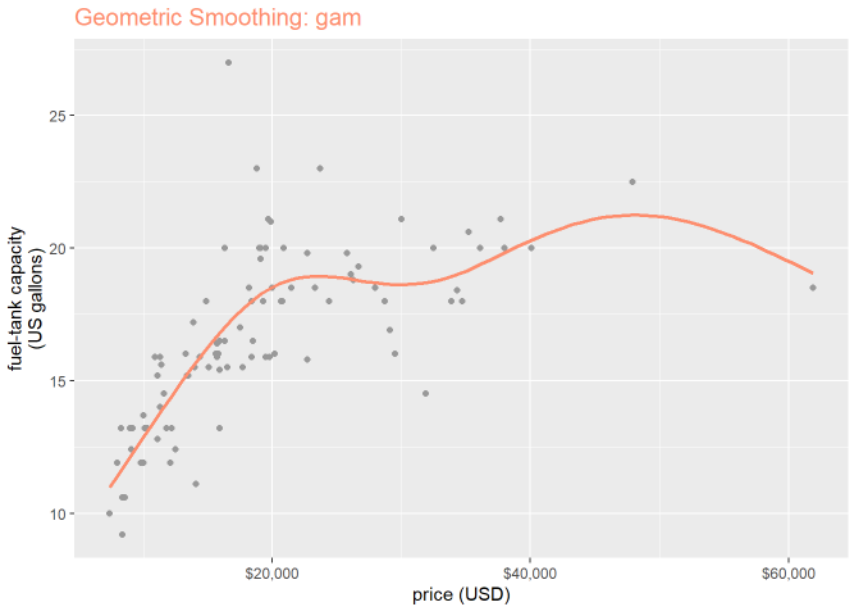


```r
#4c,d,e - lm, fe8d6d
ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
geom_point(color = "grey60") +
geom_smooth(se = FALSE, method = "lm", formula = y ~ x, color = "#fe8d6d") +
scale_x_continuous(
name = "price (USD)",
breaks = c(20, 40, 60),
labels = c("$20,000", "$40,000", "$60,000")
) +
scale_y_continuous(name = "fuel-tank capacity\n(US gallons)")+
labs(title = "Geometric Smoothing: lm") +
theme(
  plot.title = element_text(size = 14, color = "#fe8d6d"))
```

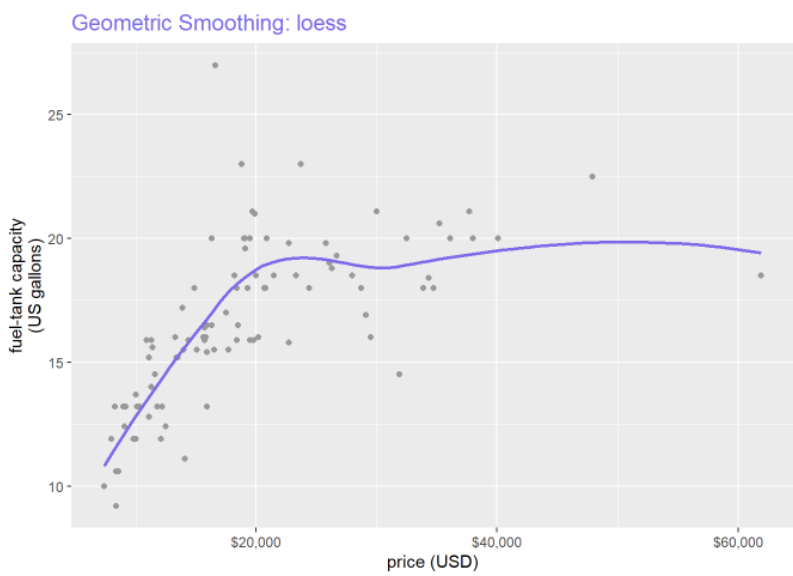



```r
#4c,d,e - glm, fe8d6d
ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
geom_point(color = "grey60") +
geom_smooth(se = FALSE, method = "glm", formula = y ~ x, color = "#fe8d6d") +
scale_x_continuous(
name = "price (USD)",
breaks = c(20, 40, 60),
labels = c("$20,000", "$40,000", "$60,000")
) +
scale_y_continuous(name = "fuel-tank capacity\n(US gallons)")+
labs(title = "Geometric Smoothing: glm") +
theme(
  plot.title = element_text(size = 14, color = "#fe8d6d"))
```

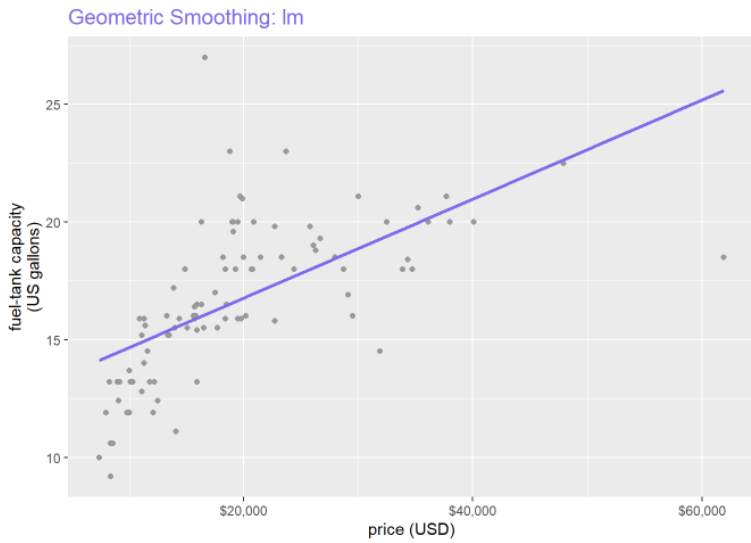
```r
#4c,d,e - gam, fe8d6d
ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
geom_point(color = "grey60") +
geom_smooth(se = FALSE, method = "gam", formula = y ~ s(x), color = "#fe8d6d") +
scale_x_continuous(
name = "price (USD)",
breaks = c(20, 40, 60),
labels = c("$20,000", "$40,000", "$60,000")
) +
scale_y_continuous(name = "fuel-tank capacity\n(US gallons)")+
labs(title = "Geometric Smoothing: gam") +
theme(
  plot.title = element_text(size = 14, color = "#fe8d6d"))
```



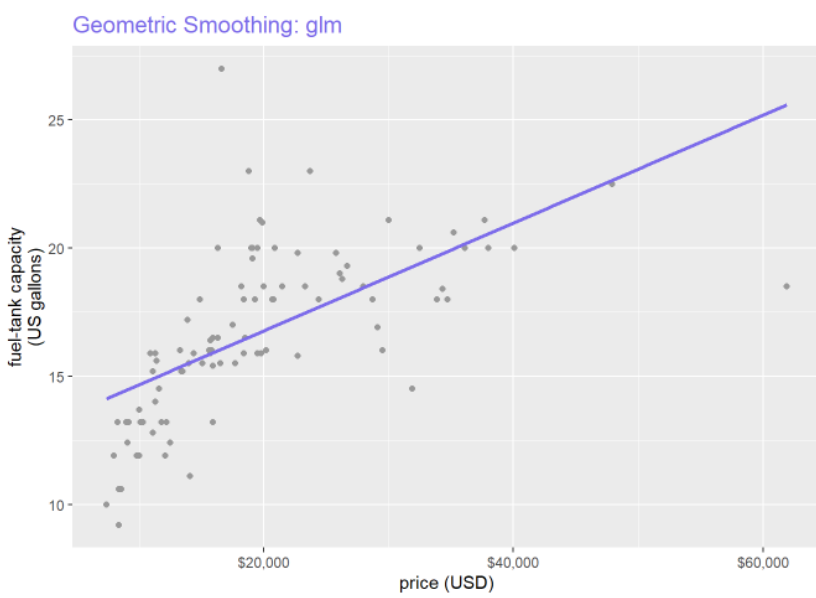

```r
#4c,d,e, 7c6bea
ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
geom_point(color = "grey60") +
geom_smooth(se = FALSE, method = "loess", formula = y ~ x, color = "#7c6bea") +
scale_x_continuous(
name = "price (USD)",
breaks = c(20, 40, 60),
labels = c("$20,000", "$40,000", "$60,000")
) +
scale_y_continuous(name = "fuel-tank capacity\n(US gallons)")+
labs(title = "Geometric Smoothing: loess") +
theme(
  plot.title = element_text(size = 14, color = "#7c6bea"))
```



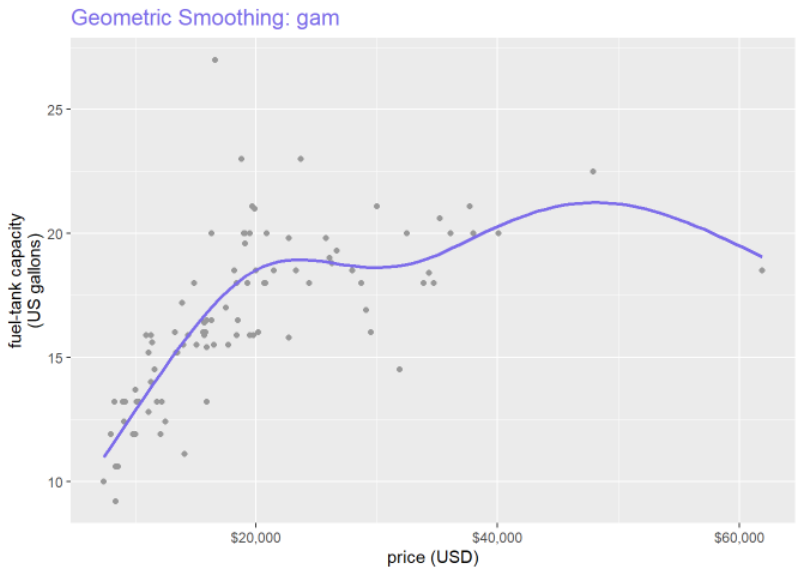
```r
#4c,d,e - lm, 7c6bea
ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
geom_point(color = "grey60") +
geom_smooth(se = FALSE, method = "lm", formula = y ~ x, color = "#7c6bea") +
scale_x_continuous(
name = "price (USD)",
breaks = c(20, 40, 60),
labels = c("$20,000", "$40,000", "$60,000")
) +
scale_y_continuous(name = "fuel-tank capacity\n(US gallons)")+
labs(title = "Geometric Smoothing: lm") +
theme(
  plot.title = element_text(size = 14, color = "#7c6bea"))
```




```r
#4c,d,e - glm, 7c6bea
ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
geom_point(color = "grey60") +
geom_smooth(se = FALSE, method = "glm", formula = y ~ x, color = "#7c6bea") +
scale_x_continuous(
name = "price (USD)",
breaks = c(20, 40, 60),
labels = c("$20,000", "$40,000", "$60,000")
) +
scale_y_continuous(name = "fuel-tank capacity\n(US gallons)")+
labs(title = "Geometric Smoothing: glm") +
theme(
  plot.title = element_text(size = 14, color = "#7c6bea"))
```



```r
#4c,d,e - gam, 7c6bea
ggplot(cars93, aes(x = Price, y = Fuel.tank.capacity)) +
geom_point(color = "grey60") +
geom_smooth(se = FALSE, method = "gam", formula = y ~ s(x), color = "#7c6bea") +
scale_x_continuous(
name = "price (USD)",
breaks = c(20, 40, 60),
labels = c("$20,000", "$40,000", "$60,000")
) +
scale_y_continuous(name = "fuel-tank capacity\n(US gallons)")+
labs(title = "Geometric Smoothing: gam") +
theme(
  plot.title = element_text(size = 14, color = "#7c6bea"))
```



```r
#5
load("G:/My Drive/Personal/Grad Program/DSC 502 - Data Visualization for EDA/R Scripts./preprint_growth.rda")

head(preprint_growth)
```

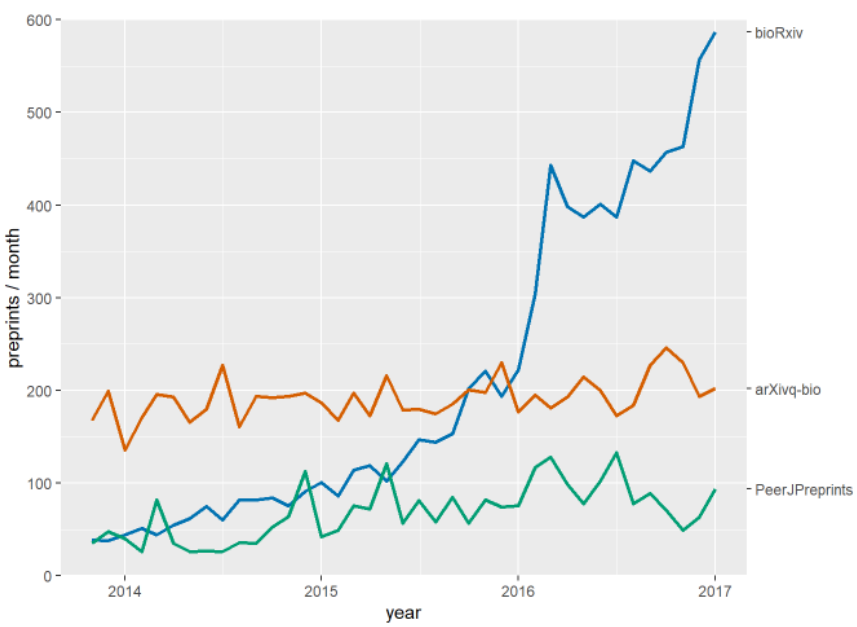
```
## # A tibble: 6 x 3
##   archive           date       count
##   <chr>             <date>     <int>
## 1 arXiv q-bio       2007-01-01    40
## 2 Nature Precedings 2007-01-01     3
## 3 F1000Research     2007-01-01     0
## 4 PeerJ Preprints   2007-01-01     0
## 5 bioRxiv           2007-01-01     0
## 6 Winnower          2007-01-01     0
```

```r
preprint_growth %>% filter(archive == "bioRxiv") %>%
filter(count > 0) -> biorxiv_growth

preprints<-preprint_growth %>% filter(archive %in%
c("bioRxiv", "arXiv q-bio", "PeerJ Preprints")) %>%filter(count > 0) %>%
mutate(archive = factor(archive, levels = c("bioRxiv", "arXiv q-bio", "PeerJ Preprints")))

preprints_final <- filter(preprints, date == ymd("2017-01-01"))

ggplot(preprints) +
aes(date, count, color = archive, fill = archive) +
geom_line(linewidth = 1) +
scale_y_continuous(
limits = c(0, 600), expand = c(0, 0),
name = "preprints / month",
sec.axis = dup_axis( #this part is for the second y axis
breaks = preprints_final$count, #and we use the counts to position our labels
labels = c("arXivq-bio", "PeerJPreprints", "bioRxiv"),
name = NULL)
) +
scale_x_date(name = "year",
limits = c(min(biorxiv_growth$date), ymd("2017-01-01"))) +
scale_color_manual(values = c("#0072b2", "#D55E00", "#009e73"),
name = NULL) +
theme(legend.position = "none")
```



```r
#5a
preprint_full = preprint_growth %>% drop_na() %>% filter(count > 0, year(date) > 2004)
head(preprint_full)
```

```
## # A tibble: 6 x 3
##   archive           date       count
##   <chr>             <date>     <int>
## 1 arXiv q-bio       2007-01-01    40
## 2 Nature Precedings 2007-01-01     3
## 3 arXiv q-bio       2007-02-01    44
## 4 arXiv q-bio       2007-03-01    55
## 5 arXiv q-bio       2007-04-01    41
## 6 arXiv q-bio       2007-05-01    59
```

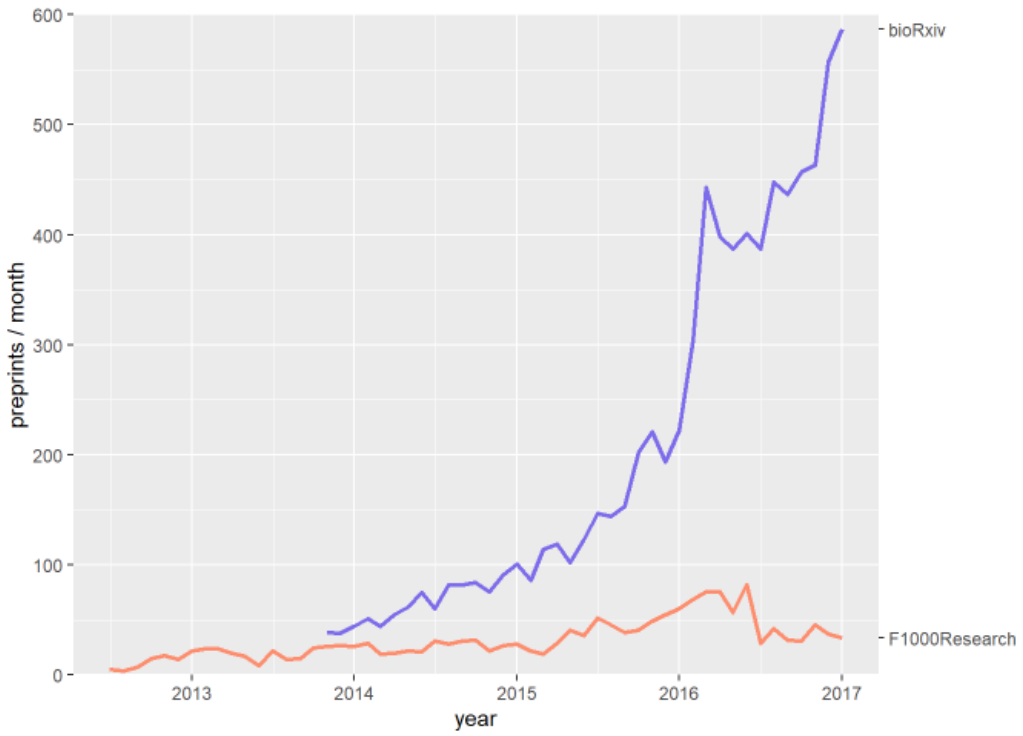
```r
#5b
preprint_filter = preprint_full %>% filter(archive %in% c("bioRxiv", "F1000Research"))
head(preprint_filter)
```

```
## # A tibble: 6 x 3
##   archive       date       count
##   <chr>         <date>     <int>
## 1 F1000Research 2012-07-01     5
## 2 F1000Research 2012-08-01     4
## 3 F1000Research 2012-09-01     7
## 4 F1000Research 2012-10-01    15
## 5 F1000Research 2012-11-01    18
## 6 F1000Research 2012-12-01    14
```

```r
preprint_final = filter(preprint_filter, date == ymd("2017-01-01"))
```

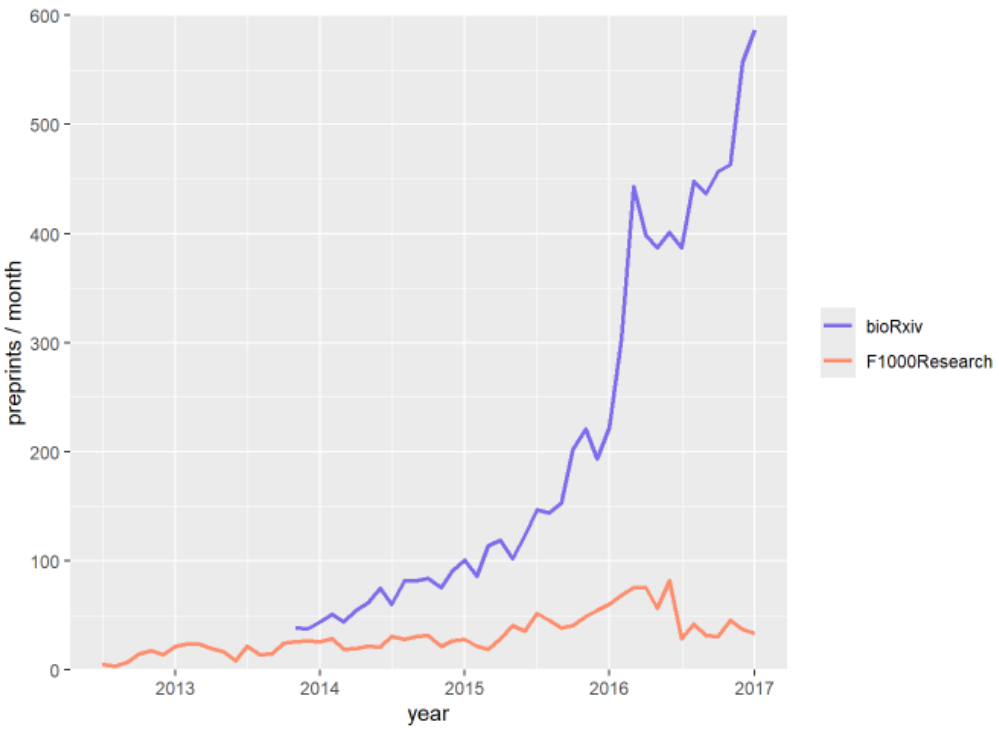
```
#5c
ggplot(preprint_filter) +
aes(date, count, color = archive, fill = archive) +
geom_line(linewidth = 1) +
scale_y_continuous(
limits = c(0, 600), expand = c(0, 0),
name = "preprints / month",
sec.axis = dup_axis(
breaks = preprint_final$count,
labels = c("F1000Research", "bioRxiv"),
name = NULL)
) +
scale_x_date(name = "year",
limits = c(min(preprint_filter$date), ymd("2017-01-01"))) +
scale_color_manual(values = c("bioRxiv" = "#7c6bea", "F1000Research" = "#fe8d6d"),
name = NULL) +
theme(legend.position = "none")
```

```
## Warning: Removed 28 rows containing missing values or values outside the scale range
## (`geom_line()`).
```



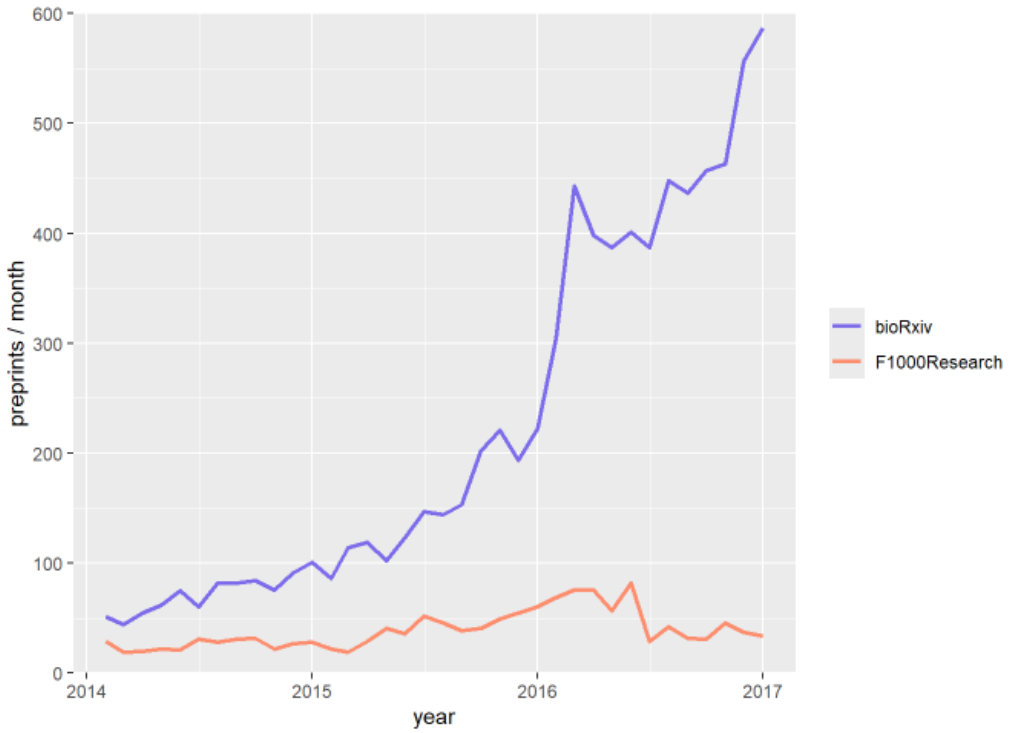
```
#5d
ggplot(preprint_filter) +
aes(date, count, color = archive, fill = archive) +
geom_line(linewidth = 1) +
scale_y_continuous(
limits = c(0, 600), expand = c(0, 0),
name = "preprints / month"
) +
scale_x_date(name = "year",
limits = c(min(preprint_filter$date), ymd("2017-01-01"))) +
scale_color_manual(values = c("bioRxiv" = "#7c6bea", "F1000Research" = "#fe8d6d"),
name = NULL) +
theme(legend.position = "right")
```

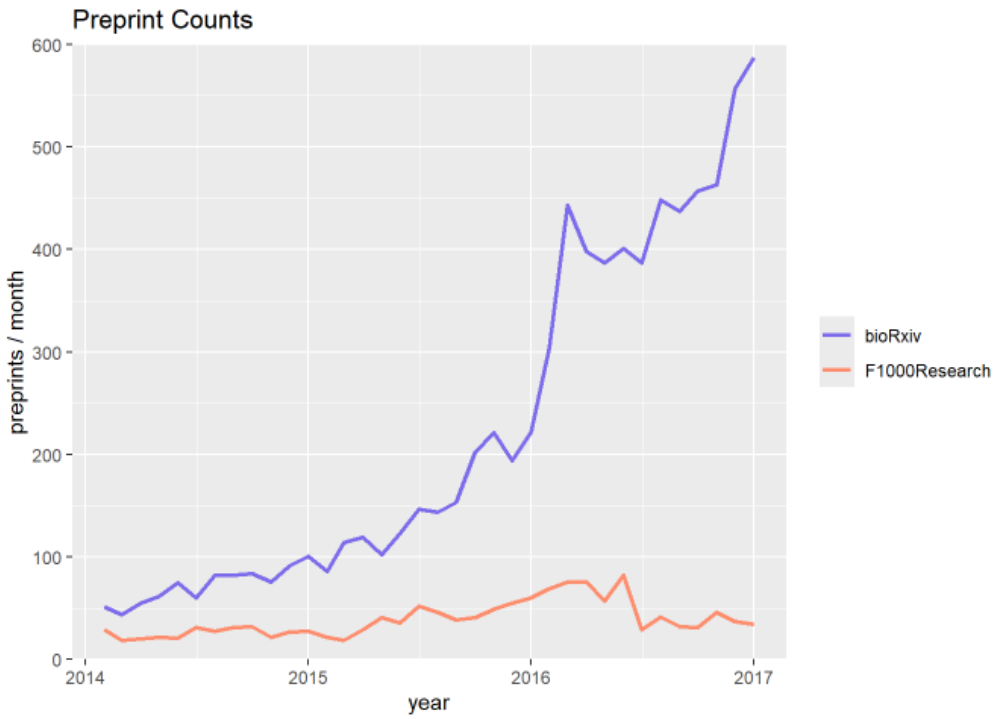
```
## Warning: Removed 28 rows containing missing values or values outside the scale range
## (`geom_line()`).
```



```
#5e
ggplot(preprint_filter) +
aes(date, count, color = archive, fill = archive) +
geom_line(linewidth = 1) +
scale_y_continuous(
limits = c(0, 600), expand = c(0, 0),
name = "preprints / month"
) +
scale_x_date(name = "year",
limits = c(ymd("2014-02-01"), ymd("2017-01-01"))) +
scale_color_manual(values = c("bioRxiv" = "#7c6bea", "F1000Research" = "#fe8d6d"),
name = NULL) +
theme(legend.position = "right")
```

```
## Warning: Removed 50 rows containing missing values or values outside the scale range
## (`geom_line()`).
```



```r
#5f
ggplot(preprint_filter) +
aes(date, count, color = archive, fill = archive) +
geom_line(linewidth = 1) +
scale_y_continuous(
limits = c(0, 600), expand = c(0, 0),
name = "preprints / month"
) +
scale_x_date(name = "year",
limits = c(ymd("2014-02-01"), ymd("2017-01-01"))) +
scale_color_manual(values = c("bioRxiv" = "#7c6bea", "F1000Research" = "#fe8d6d"),
name = NULL) +
labs(title = "Preprint Counts") +
theme(legend.position = "right")
```

```
## Warning: Removed 50 rows containing missing values or values outside the scale range
## (`geom_line()`).
```